Memory and chaos in a physics-constrained relaxation substrate: phase diagram, multi-timescale forgetting, and disturbance robustness

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Abstract

Reservoir computing exploits the fading-memory dynamics of a physical substrate, yet the memory–chaos trade-off is usually studied on abstract recurrent networks with hand-tuned leak rates. We characterize a physics-constrained relaxation substrate inspired by Si_3N_4 shallow-trap charge storage: N units with a log-normal time-constant spectrum (median $\tau_0 \approx 174\,\mu$s, coefficient of variation CV = 0.20) coupled through a per-pulse, topology-dependent modulation of the injection coefficient. Sweeping the coupling strength κ across five topology families (three negative-feedback families undergo the transition described below) and measuring the finite-time Lyapunov exponent λ, the held-out Jaeger memory capacity, and input separation, we find a sharp order–chaos transition at $\kappa^* \in (25, 30)$: the held-out memory capacity peaks 24–53% above the uncoupled baseline just before the transition, and deep chaos destroys memory. The decay of linear memory follows an analytically derived forgetting kernel $M(t) = \int p(\tau)e^{-t/\tau}\,d\tau$ over the log-normal trap spectrum (Pearson $r = 0.97$ against the measured memory-capacity curve). The $1/e$ horizon stays near $\tau_0/\bar{\Delta t} \approx 16$ pulses as the spectrum widens (numerically 12–16 pulses for CV $\in [0.02, 1.0]$), while the width CV controls the tail weight. Finally, a homeostatic regulator that estimates λ online and adjusts κ to a near-critical target improves post-disturbance held-out memory by 7.9–18% under temperature drift, edge pruning, and readout noise.

Keywords: chaos; reservoir computing; memory capacity; forgetting kernel; homeostasis; relaxation substrate.

1 Introduction

Physical reservoir computing proposes that a complex dynamical substrate — a spiking neural microcircuit [1], an optoelectronic system [2], a memristive array [3], or a charge-trap device [4] — can serve as a nonlinear, fading-memory kernel whose instantaneous state is linearly read out by a trained network. Physical implementations now span optoelectronic delay systems [2, 5], photonic integrated circuits [6], a single nonlinear node under time multiplexing [7], and neuromorphic device platforms [8, 9]; see [10, 11, 12] for reviews. The theoretical foundations are well established: the fading-memory property dates to the Volterra/Wiener system-identification literature [13], and echo state networks [14, 15] and liquid state machines [1] rely on the *echo state property* (contracting, input-forgetting dynamics), and the *edge of chaos* maximizes the computational capability of driven recurrent systems [16, 17, 18], building on the earlier characterization of the onset of chaos in

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random recurrent networks [19]. Dambre et al. [20] formalized the total information-processing capacity of dynamical systems, showing a universal trade-off between linear memory and nonlinear processing. On the memory side, multi-timescale synaptic models [21, 22] show that memory retention is dramatically extended by a *spectrum* of time constants: cascade models achieve near power-law forgetting from many exponentially decaying components.

For a physical reservoir, the time-constant spectrum is not a free hyperparameter — it is a *material property*. Shallow-trap charge storage in Si_3N_4 exhibits thermally activated relaxation with a log-normal spread of trap time constants; our prior work [4] established that this substrate encodes pulse-interval patterns through exponential relaxation with $\tau_0 \approx 174\,\mu$s at 300 K and a device-to-device spread of CV = 0.20, achieving 79.6% four-class hybrid encoding accuracy in a purely numerical setting (Digital-Model: the calibrated device simulator, with no fabricated hardware in the loop; all experiments in this paper are at this same fidelity level). That work, however, treated the substrate as a *parallel* array of independent devices; the coupling between devices — the element that could push the substrate toward richer, near-critical dynamics — was only implemented as a quasi-static modulation of the effective injection coefficient across blocks.

This paper closes that gap. We introduce the *per-pulse temporal generalization* of the topology coupling: every pulse, each unit's injection coefficient is modulated by a topology-dependent contrast against its neighbors' current ratios, making the substrate a genuinely recurrent, tunable dynamical system with a physical chaos knob κ. Our contributions are:

1. **A memory–chaos phase diagram** for a physics-constrained relaxation substrate: the finite-time Lyapunov exponent λ, the held-out Jaeger memory capacity, and the input separation, all as functions of the coupling strength κ, across five topology families (bidirectional ring, unidirectional ring, hub_star, lateral-inhibition ring, random graph) and two coupling sign conventions (negative- and positive-feedback contrast), with an additive state-coupling control.

2. **An analytic forgetting kernel** $M(t) = \int p(\tau)e^{-t/\tau}\,d\tau$ for the log-normal trap spectrum, validated against the measured memory-capacity curve (Pearson $r = 0.97$), which turns the device-physics spread CV into a *design knob for the forgetting curve.*

3. **A homeostatic chaos regulator** that estimates λ online from Benettin twin trajectories and adjusts κ to a near-critical target, restoring 7.9–18% of held-out memory after environmental disturbances (temperature drift, edge pruning, readout noise) where fixed coupling cannot.

Throughout, we use a *held-out* memory-capacity protocol: ridge fits are trained on the first 70% of a driving stream and evaluated on the last 30% (with a $k_{\max}$-length leakage buffer). We show that the common in-sample Jaeger convention dramatically overstates memory in the chaotic regime, which is a methodological caution for the field.

2 Substrate model

2.1 Single-unit relaxation

Each unit models a Si_3N_4 shallow-trap population — trap-controlled charge transport of this kind is documented independently in SiN_x memristors [23] — with effective trap occupancy $x_i \in [0, 1]$. A write pulse of width $p_w = 1\,\mu$s injects charge with coefficient α; the occupancy then relaxes exponentially with the unit's trap time constant τ_i both during the pulse width and during the inter-pulse interval Δt_t. The per-pulse update is

$$x_i(t+1) = \Big[x_i(t) + \alpha_{\mathrm{eff},i}(t)\big(1 - x_i(t)\big)\Big]\, e^{-p_w/\tau_i} e^{-\Delta t_t/\tau_i}, \qquad x_i \in [0, 1], \tag{1}$$

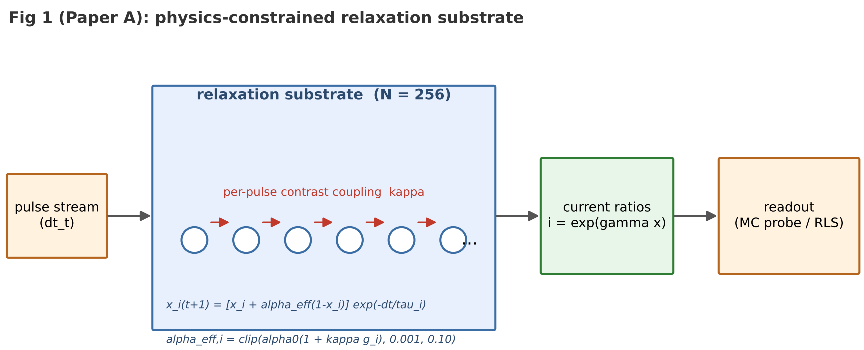

Figure 1: Substrate schematic: pulse stream, $N = 256$ relaxation units with log-normal τ spectrum, per-pulse contrast coupling (red arrows, κ), and the readout probe.

and the observable current ratio is $i_i = e^{\gamma x_i}$ with $\gamma = \ln 100$ (a 100:1 ON/OFF ratio at $I_{\mathrm{HRS}} = 50$ fA). The trap time constants are drawn from a log-normal distribution with median $\tau_0 = 174\,\mu$s (from $E_a = 0.55$ eV, $\nu = 10^{13}\,\mathrm{s}^{-1}$, $T = 300$ K) and CV = 0.20: $\tau_i \sim \mathrm{Lognormal}(\mu, \sigma)$ with $\sigma^2 = \ln(1+\mathrm{CV}^2)$; the log-normal form follows from a Gaussian spread of activation energies (Appendix A.1).

2.2 Per-pulse topology coupling

The injection coefficient of unit i at pulse t is modulated by a topology-dependent contrast between its own current ratio and a weighted mean of its neighbors' current ratios:

$$\alpha_{\mathrm{eff},i}(t) = \mathrm{clip}\Big(\alpha_0\big(1 + \kappa\, g_i(t)\big),\ \alpha_{\min}, \alpha_{\max}\Big), \tag{2}$$

with two contrast conventions — *negative feedback* (mode 1): $g_i = (\bar{i}_{\mathcal{N}(i)} - i_i)/i_i$; and *positive feedback* (mode 2): $g_i = (i_i - \bar{i}_{\mathcal{N}(i)})/\bar{i}_{\mathcal{N}(i)}$ — where $\bar{i}_{\mathcal{N}(i)}$ is the weighted mean of neighbor current ratios (row-normalized weights). The base injection coefficient is $\alpha_0 = 0.02$; the physical clip range $[\alpha_{\min}, \alpha_{\max}] = [0.001, 0.10]$ bounds the injection coefficient. A control condition replaces the contrast coupling with an additive state drive $x_i \mathrel{+}= \kappa\, \bar{x}_{\mathcal{N}(i)}$ after the standard update (the non-physical, "textbook ESN" style). All coupling is computed two-pass (contrasts from the pre-update state, then all units updated), making the step order-independent (Appendix A.2). At $\kappa = 0$ the model reduces exactly to the parallel array of [4] (verified to machine precision). Fig. 1 summarizes the full pipeline: the pulse stream, the relaxation array with its log-normal τ spectrum, the per-pulse contrast coupling, and the linear readout probe.

Five topologies over the $N = 256$ units are used: **ring_bidir** (each unit watches its two neighbors), **ring_unidir** (each unit watches its predecessor), **hub_star** (satellites watch a fixed hub), **lateral_ring** (distance-decayed neighbors within radius 4), and **random_graph** (Erdős–Rényi, mean degree 8, fixed structure across the sweep; graph-to-graph variability of this fixed-instance choice is quantified in Section 4.1). The drive is an i.i.d. uniform interval stream $\Delta t_t \sim U[2, 20]\,\mu$s, the fast-drive regime in which the nominal memory window is $N_{\mathrm{eff}} = \tau_0/\bar{\Delta t} \approx 16$ pulses. This i.i.d.

drive differs from the prior parallel-array benchmark protocol, which keyed class information into two *fixed* intervals (20 and 200 μs) [4]; the interval notation of the two protocols is unrelated.

3 Characterization methods

Finite-time Lyapunov exponent. For a driven system the relevant quantity is sensitivity to initial conditions under identical drive. We evolve a twin trajectory perturbed by $\epsilon = 10^{-8}$ per unit, renormalize the separation to $\epsilon\sqrt{N}$ every 10 pulses (Benettin scheme [24]), and report $\lambda = \langle \ln(d/\epsilon\sqrt{N}) \rangle$ per pulse, averaged over the full stream (detailed iteration steps in Appendix A.4).

Held-out memory capacity. Following Jaeger, the linear memory of lag k is the squared Pearson correlation between the ridge-decoded value of the driving interval Δt_{t-k} and its true value, decoded from the current-ratio observables at time t. We fit the ridge readout [25] (regularization $\lambda_r = 1$, features standardized with training-segment statistics) on the first 70% of the collected stream, hold out a $k_{\max} = 50$-step leakage buffer, and evaluate on the last 30%; $\mathrm{MC}_{\mathrm{total}} = \sum_{k=1}^{50} r_k^2$. The held-out protocol is essential: the in-sample convention inflates memory in the chaotic regime by 7–19$\times$ relative to held-out evaluation (Section 4.4). The estimator derivation and the linear-reservoir closed form that anchors the theory are given in Appendix A.5.

Input separation. The RMS distance between the state trajectories produced by two different i.i.d. drives from the same initial condition, averaged over the collected window — a proxy for how far different input histories push the state.

Activity proxy. The running standard deviation of the population-mean current ratio over a 200-pulse window. It is not reported as a result below; it enters only as the quantity on which the fixed-activity-target baseline of Section 6 is expressed.

Reproducibility details. All phase-diagram runs use $N = 256$ units, CV = 0.20, and 10 independent seeds; the τ draws are *paired* across κ within a topology (the same seed index maps to the same τ vector at every κ, so κ acts as the dose and the substrate draw as the paired block), and the random-graph structure is fixed across the entire sweep. Every trajectory starts from $x_i = 0$ and is *preprogrammed* with 50 write pulses (width 1 μs, interval 2 μs) at the base injection coefficient $\alpha_0 = 0.02$, which sets the initial occupancy distribution; the first 200 pulses of each 1200-pulse trajectory are discarded as washout before any statistics are collected. The FTLE twin uses $\epsilon = 10^{-8}$ per unit with renormalization every 10 pulses. The disturbance experiments of Section 6 use the same substrate (10 seeds) driven by a discretized Mackey–Glass series ($x_{t+1} = (1-b)x_t + a\,x_{t-\tau}/(1+x_{t-\tau}^{10})$ with $a = 0.2$, $b = 0.1$, $\tau_{\mathrm{MG}} = 17$ steps; 300-step warmup discarded; values normalized and mapped monotonically onto intervals in $[2, 20]\,\mu$s; the target is the next series value), read out by an online recursive least squares rule (forgetting factor 0.999, initial covariance 1, regularization 10^{-4}, $N+1$ features). The homeostat updates every 1000 pulses from a 400-pulse Benettin pair, with feedback gain $\eta = 3$ and regulatory bounds $\kappa \in [1, 60]$.

4 Phase diagram

All quantities below are means over 10 independent seeds (paired τ draws across κ); the full table is committed as data/substrate_phase_diagram_v2.csv. Table 1 summarizes the per-topology behavior, and Fig. 2 traces the full κ-dependence of the finite-time Lyapunov exponent, the held-out and in-sample MC, and the clip fraction across the topology families.

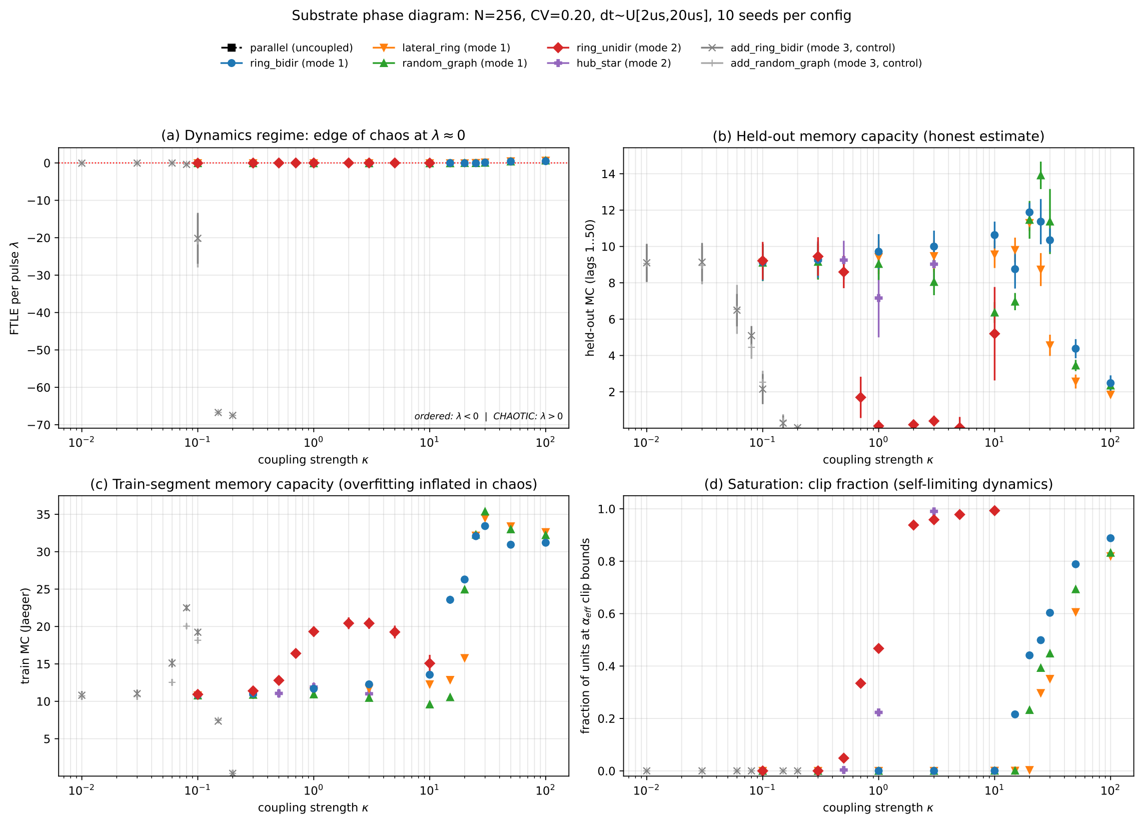

Figure 2: Phase diagram: κ vs. finite-time Lyapunov exponent, held-out and in-sample MC, and clip fraction across topology families.

4.1 Negative-feedback family: stabilization then a sharp transition

For the mode-1 (negative-feedback) topologies, weak-to-moderate coupling *stabilizes*: λ becomes more negative ($-0.065 \to -0.090$ as κ grows to 10 for the ring) and the mean absolute contrast $|g|$ shrinks ($0.15 \to 0.01$), i.e., the coupling actively homogenizes neighboring states. Memory rises modestly in this regime (ring_bidir held-out MC $9.1 \to 10.6$ at $\kappa = 10$). At $\kappa \in (25, 30)$ the system crosses a sharp order–chaos transition (λ jumps from ≈ -0.05 to > 0) for all three mode-1 topologies; the transition is accompanied by the onset of α_{eff} clipping (clip fraction 0.3–0.6), i.e., the physical bounds start to bind. **The held-out memory capacity peaks in the last ordered configuration before the transition** (peak positions are quoted at the coarse grid resolution $\Delta\kappa = 5$; the fine sweep of Section 4.2 confirms that each peak precedes its topology's crossing): random_graph at $\kappa = 25$ reaches MC $= 13.91$ vs. 9.07 uncoupled (+53%), ring_bidir at $\kappa = 20$ +30%, lateral_ring at $\kappa = 20$ +24%. The fixed-instance design keeps graph-sample noise out of the topology comparison; a follow-up sweep over 9 independent Erdős–Rényi instances (data/s36_topo_instance_variability_v1.csv) bounds it: peak held-out MC 13.47 ± 0.57 (range 12.80–14.35), i.e. $+48.5\% \pm 6.3\%$ over uncoupled, with the peak at $\kappa = 25$ (7/9 instances) or $\kappa = 30$ (2/9); the fixed instance of Table 1 (13.91, +53%) lies inside this envelope.

Table 1: Per-topology summary of the κ sweep (held-out MC; mean $\pm$ s.d. over 10 seeds). Peak gains are relative to the uncoupled baseline MC $= 9.07 \pm 1.65$.

Topology	Mode	Transition / behavior	Peak MC (vs. 9.07)
ring_bidir	negative	order–chaos at $\kappa^* \in (25, 30)$	11.88 ± 0.84 (+30%) at $\kappa = 20$
lateral_ring	negative	order–chaos at $\kappa^* \in (25, 30)$	11.29 ± 0.70 (+24%) at $\kappa = 20$
random_graph	negative	order–chaos at $\kappa^* \in (25, 30)$	13.91 ± 1.22 (+53%) at $\kappa = 25$
ring_unidir	positive	self-limited criticality ($\lambda \approx -0.002$), $\kappa \in [1, 5]$	0.12–0.39*
hub_star	positive	frozen (clip fraction 1.0)	none

*no held-out linear memory (in-sample MC 18–20).

4.2 Transition mechanism: coupling versus the physical clip

The sharp transition at κ^* coincides with the onset of α_{eff} clipping (clip fraction 0.3–0.6), which raises the question of whether the chaos is produced by the topology coupling or by the physical bounds. A follow-up sweep (data/s27_clip_kappa _fine_v1.csv; $\kappa \in [20, 30]$ at unit resolution $\times$ $\alpha_{\max} \in \{0.1, 0.2, 0.5\}$, 10 seeds) answers it. First, the fine grid pins the mean-FTLE crossing at $\kappa^* = 25.3$ (lateral_ring), 27.4 (ring_bidir), and 27.9 (random_graph), all inside the $(25, 30)$ bracket above — the transition is sharp at unit resolution, not a grid artifact. Second, widening the clip five-fold ($\alpha_{\max} = 0.5$) does not move the transition for the two topologies with the largest memory gains: $\kappa^* = 25.3 \to 25.4$ (lateral_ring; shift +0.05) and $27.9 \to 27.7$ (random_graph; shift -0.21). The order–chaos transition is therefore *coupling-driven*: the clip is a consequence of the large contrasts it bounds, not their cause. ring_bidir is the one exception — its crossing moves +3.4 under the wider clip, so the bound contributes to that topology's transition onset. A secondary result: widening the clip raises the ring_bidir memory peak from 11.9 to 17.7 ($\alpha_{\max} = 0.2$), showing that for ring-like topologies the physical clip currently caps held-out memory — a fabrication-tunable trade.

4.3 Positive-feedback family: self-limited criticality, separation without linear memory

The mode-2 (positive-feedback) topologies behave very differently. The **unidirectional ring** self-limits at criticality: λ approaches ≈ -0.002 (within our FTLE resolution of zero) over a half-decade of coupling strength $\kappa \in [1, 5]$, held there by the α_{eff} clip (clip fraction 0.5–1.0) interacting with the $(1 - x)$ saturation; beyond $\kappa = 5$ the ordered regime reasserts itself ($\lambda \approx -0.026$ at $\kappa = 10$). This regime shows the *largest* input separation of the entire study (inter-stream RMS 0.054–0.085 vs. 0.022 baseline) but *zero* held-out linear memory (MC 0.12–0.39, despite in-sample MC of 18–20). The exponential current nonlinearity makes the saturated critical states information-rich but linearly undecodable — a clean demonstration of the separation–memory trade-off: separation-rich critical states require nonlinear readout to be exploited. The **hub_star** topology freezes instead: with no feedback loop through the hub, satellites saturate into bistable above/below-hub states (clip fraction 1.0) and memory never exceeds the baseline.

4.4 Chaos destroys memory; in-sample MC is misleading

In the chaotic regime ($\kappa = 50, 100$), the held-out MC collapses to 1.85–4.4 (a 3–7× drop from the peak) — a sharp contrast to the successful reservoir prediction of externally driven chaotic systems [26, 27], where the drive, not the internal dynamics, supplies the input trace. The in-sample (train-segment) MC, by contrast, *explodes* to 30–35 — an overfitting artifact of ridge decoding the huge-amplitude chaotic state; the held-out protocol reveals the true destruction of memory. We therefore report held-out MC throughout and recommend the practice to the field.

4.5 The additive control: why physics constraints matter

The additive state-coupling control (mode 3, textbook ESN style) reaches its best in-sample MC (17–22) near its saturation threshold ($\kappa \approx 0.08$–0.1) and then dies: the state pins at the saturation boundary and the FTLE estimate collapses numerically (a signature of degenerate dynamics). Its held-out MC never exceeds the uncoupled baseline meaningfully (max 9.1), and the operating window is a narrow band before saturation. The physics-constrained contrast coupling, by contrast, remains well-behaved across three decades of κ and achieves the memory gains above — the physical clip acts as a built-in stability regulator.

Validity domain of κ. The phase structure above is established on the swept grids — $\kappa \in [0.1, 100]$ for the contrast-coupled families and $\kappa \in [0.01, 0.2]$ for the additive control — at a single clip range $[\alpha_{\min}, \alpha_{\max}] = [0.001, 0.10]$ and CV $= 0.20$; the clip-ablation sweep of Section 4.2 additionally covers $\kappa \in [20, 30]$ at unit resolution for $\alpha_{\max} \in \{0.1, 0.2, 0.5\}$, and the regulator of Section 6 operates within the regulatory bounds $\kappa \in [1, 60]$. Between coarse grid points the transition location is resolved only to its bracketing interval, and outside the swept or clipped ranges — or for spectrum parameters other than those swept in Supplementary Note 1 — the diagram should not be extrapolated.

5 Multi-timescale forgetting theory

5.1 The kernel

The linear memory of a single trap for an input at lag t decays as $e^{-t/\tau}$. The substrate's population forgetting kernel is the average over the log-normal spectrum:

$$M(t) = \int_0^\infty p(\tau)\, e^{-t/\tau}\, d\tau, \qquad p(\tau) = \mathrm{Lognormal}(\tau_0, \mathrm{CV}). \tag{3}$$

We evaluate $M(t)$ by Gauss–Hermite quadrature (160 nodes, exact to machine precision; derivation in Appendix A.3). Three properties follow (Fig. 3):

1. **The $1/e$ horizon is median-pinned.** $M(\tau_0/\bar{\Delta} t) \approx e^{-1}$ — within a few percent for CV ≤ 0.5 and within $\sim$25% at CV $= 1.0$; the horizon is 12–16 pulses for CV $\in [0.02, 1.0]$, in agreement with the nominal window $N_{\text{eff}} \approx 16$. The *median* time constant sets where memory is; the width sets how it decays.

2. **The tail steepens with narrowing spectrum.** The log-log slope $d \ln M / d \ln t$, fitted over the long-lag window $t \in [5{,}000,\ 20{,}000]$ pulses (Appendix A.3), is -60.6 at CV $= 0.2$ and rises to -7.1 at CV $= 1.0$ (and $-\infty$ in the CV$\to 0$ single-exponential limit). A wider trap spread yields a heavier tail — slower forgetting of old information.

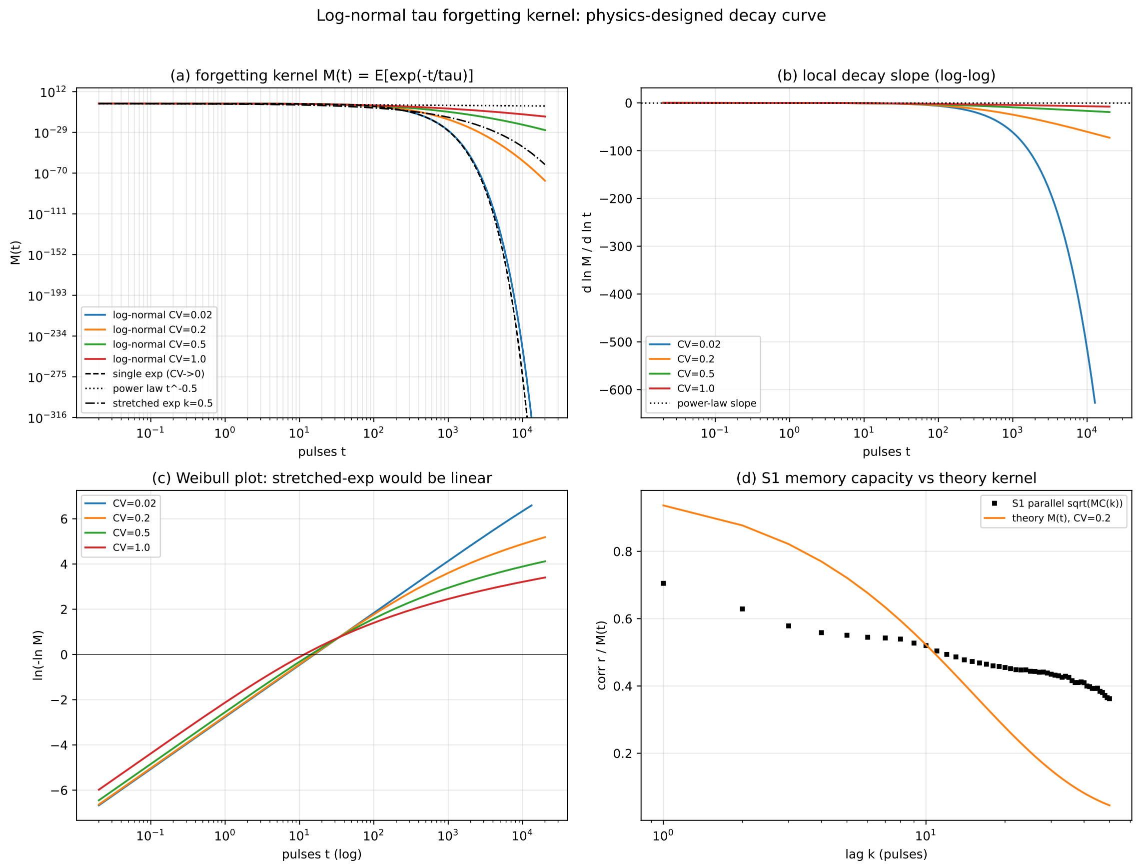

Figure 3: Forgetting kernel: $M(t)$ curves over CV, tail slopes, and the overlay of the measured $\sqrt{\mathrm{MC}(k)}$ on $M(k\bar{\Delta}t)$ (Pearson $r = 0.97$).

3. **Comparison to cascade ideals.** The log-Gaussian tail $\ln M(t) \sim -(\ln t - \mu)^2/(2\sigma^2)$ is *slower* than a single exponential but *steeper* than the near power-law retention of multi-timescale cascade models [21, 22]. Reaching power-law-like retention would require an unrealistically wide spectrum; the physical substrate therefore occupies a specific point in the forgetting design space set by CV.

5.2 Validation against the measured memory capacity

The measured parallel-substrate curve $\sqrt{\mathrm{MC}(k)}$ (held-out) tracks $M(k \cdot \bar{\Delta}t)$ at CV $= 0.20$ with **Pearson** $r = 0.97$ (data/forgetting_curve_theory_overlay_v1.json); at lag 10 the two agree almost exactly (0.520 vs. 0.523), with systematic damping at short lags (readout regression attenuation for a random drive). The committed per-lag curves quantify the residuals: the measured curve sits ≈20% below the kernel for lags $k \leq 9$ (the readout damping), crosses it at $k \approx 10$, and sits above it at long lags (measured 0.36–0.42 vs. kernel 0.04–0.10 for $k = 36$–50), so the tail of the driven, readout-limited curve is heavier than the single-unit kernel and the kernel should be read as the bulk, not the extreme tail, of the memory. The device-physics spectrum thus predicts the substrate's empirical memory curve within the model: the kernel and the measured curve share the same log-normal spectrum assumption, so this is a model-internal consistency check rather than a prediction against independent measurement.

Implication. CV — a fabrication-controllable spread parameter — is a *design knob for the forgetting curve*. This is the quantitative basis for the "physics as a structured state-space kernel" view: the substrate realizes a specific multi-timescale kernel with a tunable tail.

The coupled-substrate caveat (task-level CV sweep). The kernel theory is single-unit; at the task level the CV knob's direction is operating-regime-dependent — uncoupled operation follows the theory (wider CV, heavier tail, more retained memory), while coupled near-critical operation prefers narrow spectra. The full sweep (CV $\in \{0.1, 0.2, 0.4\} \times$ {uncoupled, random_graph $\kappa = 25$, ring_bidir $\kappa = 20$}, 10 seeds) is given in Supplementary Note 1 (Fig. S1); the kernel-level stability across CV $\in [0.02, 1.0]$ is in Fig. 3 and Appendix A.3.

The kernel survives coupling (shape stability). The single-unit theory was validated on the parallel substrate; the committed per-lag memory curves let us ask whether the coupled substrate's curve is still kernel-shaped. The *physical* spectrum (174 μs, CV 0.20) achieves Pearson $r = 0.91$–0.99 against $\sqrt{\mathrm{MC}(k)}$ over $\kappa \in \{15, \ldots, 30\}$ of all three mode-1 topologies ($r = 0.98$ at random_graph κ=25), without any re-fitting (data/kernel_coupling_shape_v1.json; quadrature as in Appendix A.3): the kernel shape survives coupling. The approach to the transition dilates the memory kernel (consistent with the near-critical slowdown of Section 4.1), but single curves cannot separate an effective-median shift from a width change (the fit saturates at wide CV), so we report the shape stability quantitatively and the dilation qualitatively rather than as a precise renormalization map.

6 Disturbance robustness and the λ-homeostat

6.1 The disturbed memory landscape

We subject the random-graph substrate ($\kappa = 25$, the nominal optimum) to three abrupt disturbances at $t = 10$k in an online Mackey–Glass forecasting task [28] (recursive least squares, RLS, readout): **temperature drift** (all τ scaled by 1.5, i.e. a cooling drift $\approx 300 \to 294$ K under the Arrhenius model; heating would shorten τ), **edge pruning** (40% of coupling edges removed), and **readout noise** ($\sigma = 0.1\times$ feature std). The held-out MC–κ landscape shifts with the disturbance (means $\pm$ s.d. over 10 seeds): noise reduces memory everywhere (peak 10.05 ± 0.81 vs. 13.98 ± 2.05 nominal), edge pruning *improves* memory (16.53 ± 2.22 — removing homogenizing edges), and temperature drift flattens the landscape, retaining substantial memory at larger κ (11.01 ± 1.76 at $\kappa = 40$ vs. 6.95 ± 0.91 nominal). (The nominal 13.98 is the disturbance-free peak under this section's online task protocol; it is distinct from the phase-diagram peak 13.91 of Table 1, which uses the offline ridge protocol of Section 3 on the i.i.d. drive — the 0.5% offset between the two lies well within the seed-level spread of either estimate.) Consequently, any regulator anchored to a *fixed target on the activity proxy of Section 3* (a common homeostatic design) fails: the target's optimum is not disturbance-invariant (this negative design result motivated the λ-homeostat).

6.2 The λ-homeostat

The homeostat estimates λ directly: every 1000 pulses it runs a 400-pulse Benettin pair at the current κ (40% computational overhead) and steps

$$\kappa \leftarrow \mathrm{clip}\big(\kappa + \eta\, \mathrm{clip}(\lambda_{\mathrm{target}} - \hat{\lambda}, -1, 1), [1, 60]\big), \qquad \lambda_{\mathrm{target}} = -0.02, \quad \eta = 3, \tag{4}$$

the slightly-ordered side of the transition where held-out MC peaks (Section 4.1). We retain $\lambda_{\mathrm{target}} = -0.02$ in the main results as a deliberately conservative, manually tuned choice made *before* the systematic sweep below; the sweep identifies $\lambda_{\mathrm{target}} = 0$ as optimal, so the main-result gains

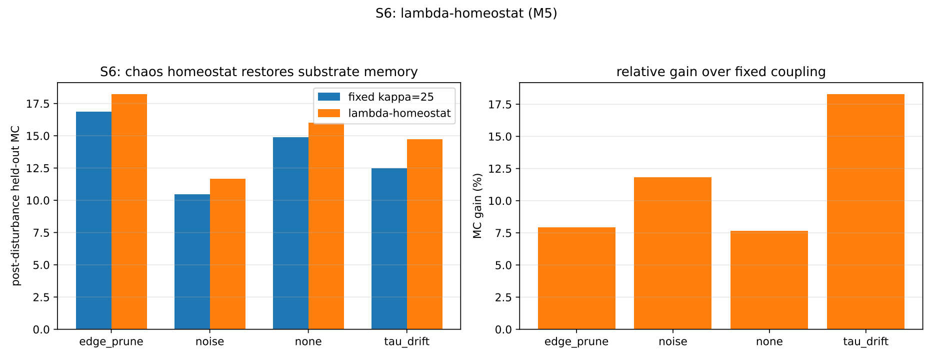

Figure 4: λ-homeostat robustness: post-disturbance held-out MC gains over fixed coupling (Table 2).

should be read as a lower bound on what a retuned target achieves. The regulated system settles at $\kappa \approx 26$–27 under all conditions and achieves post-disturbance held-out MC gains of +7.6% (none), +18% (τ-drift), +7.9% (edge pruning), +11.8% (noise) over the fixed coupling (Table 2; Fig. 4), at broadly comparable task-level normalized mean squared error (NMSE; regulated up to 2× worse on the gain arms, better under readout noise; the RLS readout absorbs task-level differences, and the substrate-level memory improvement is revealed by the readout-independent MC probe). The λ-homeostat is disturbance-type-agnostic because it tracks the dynamical quantity itself rather than a proxy whose optimum shifts.

The estimate is also robust to its own noise: injecting Gaussian error of up to $\sigma_\lambda = 0.10$ (the typical estimate magnitude is 0.05–0.1) into every Benettin estimate leaves the settled coupling essentially unchanged (settled κ at the third disturbance round r3: 28.5–28.7, under the S11-chain protocol, which differs from the Mackey–Glass task values of Table 2) and the post-disturbance r3 MC within 0.16 of the noiseless value with no significant paired difference at any noise level (two-sided paired t-test across seeds, $\alpha = 0.05$; data/s32_ftle_noise_v1.csv) — the clipped proportional feedback averages out the estimation error over the 1000-pulse interval between updates.

A systematic λ_{target} sweep ($\lambda_{\text{target}} \in \{-0.05, -0.02, -0.005, 0\} \times \text{CV} \in \{0.1, 0.2, 0.4\}$; 5 seeds, trading per-cell precision for coverage of the 12-cell grid, whereas the single-configuration runs elsewhere in this paper use 10 seeds) was executed under the companion Paper B protocol and is reported as its Table 2 [29] — the values are quoted from that work and are not re-collected under this paper's phase-diagram protocol. The sweep identifies λ_{target}=0 (the edge of chaos) as optimal: post-disturbance MC peaks near the edge of chaos at every tested CV (λ_{target}=0 optimal at CV=0.1 and 0.4; the -0.005 setting marginally ahead at CV=0.2), with gains of +25% (CV=0.1), +19% (CV=0.2), +5% (CV=0.4) over fixed coupling — updating the manually tuned -0.02 used in the main results.

Hyperparameter provenance. The parameters that enter these experiments are of three kinds: measured material properties (τ_0, CV); quantities swept in this paper (κ across three decades, Section 4; the clip range, Section 4.2; the task-level CV, Supplementary Note 1); or fixed protocol values inherited from the calibrated prior substrate and the readout conventions (α_0, the clip range, the preprogramming protocol; ridge $\lambda_r = 1$ and the 70/30 split; $\eta = 3$ and the 1000-pulse update period of the homeostat, whose robustness to estimate noise is verified above). No readout or regulator hyperparameter was tuned per condition.

Table 2: Post-disturbance held-out MC of the λ-homeostat (random_graph substrate, Mackey–Glass online task; mean $\pm$ s.d. over 10 seeds). Fixed: regulator disabled, $\kappa = 25$; regulated: settled κ (26–27).

Disturbance	Fixed MC	Regulated MC	Gain
none	14.87 ± 2.43	16.01 ± 2.38	+7.6%
τ-drift	12.46 ± 1.74	14.73 ± 1.88	+18%
edge pruning	16.87 ± 1.54	18.21 ± 1.29	+7.9%
readout noise	10.45 ± 0.99	11.68 ± 1.12	+11.8%

7 Discussion

The physical parameter range bounds the accessible computation. Three material/design quantities jointly define the substrate's computational envelope: the clip range $[\alpha_{\min}, \alpha_{\max}]$ (which bounds coupling and, in the positive-feedback family, self-limits the system at criticality), the coupling threshold κ^* (which separates the memory-rich ordered side from the memory-destructive chaotic side), and the trap spectrum (which fixes both the $1/e$ memory horizon and the forgetting tail). A clip-ablation sweep (Section 4.2) shows that κ^* is invariant to a five-fold widening of the clip for the two highest-gain topologies (random graph and lateral ring), so the threshold is a genuine property of the coupling dynamics rather than a clip artifact; the exception is the bidirectional ring, whose transition shifts by +3.4 under clip widening (Section 4.2), so the clip's role is secondary and topology-dependent. For the neuromorphic-design question of what this device can compute, the answer is a bounded but substantial region of the memory–chaos plane, fully characterized here.

Separation without linear memory. The positive-feedback family reaches true criticality ($|\lambda| \approx 0$) with maximal separation yet zero held-out linear memory. This is a caution for the common practice of tuning reservoirs to the edge of chaos: at criticality the *linear* readout loses the input trace, and the separation must be exploited nonlinearly (a companion work develops structure plasticity from functional connectivity [29]). For linear readouts, the optimal operating point is the *last ordered configuration before the transition* — a slightly-subcritical target ($\kappa = 20$–25, $\lambda \approx -0.05$), consistent with where the homeostat parks the system ($\kappa \approx 26$–27).

The forgetting kernel as a bridge between device physics and state-space models. $M(t)$ connects device physics to the state-space-model literature: the substrate is a hardware-realizable structured state-space kernel whose decay spectrum is set by fabrication (CV). The $r = 0.97$ validation shows the connection is quantitative, not metaphorical.

Spectrum knobs: location versus width. Across the companion papers three single-parameter spectrum knobs appear — the material CV here, the algorithmic EMA timescale τ_m (companion Paper C [30]), and the log-uniform coverage range of the state-space host (companion Paper D [31]). They are not interchangeable, and the distinction is quantitative: for a log-normal spectrum the $1/e$ horizon is set by the *location* (median τ_0: $M(\tau_0) \approx e^{-1}$ — within a few percent for CV ≤ 0.5, within $\sim$25% at CV $= 1.0$; numerically 12–16 pulses over CV $\in [0.02, 1.0]$) while the tail weight is set by the *width* (log-log slope -60.6 at CV 0.2 vs. -7.1 at CV 1.0).

Paper C's τ_m is a location knob (the synthesized kernel's horizon scales with τ_m, tail shape fixed), and the log-uniform range is a width knob of a different family. The practical corollary: specifying both a horizon and a tail requires two independent knobs, and single-knob tuning moves exactly one observable cross-section — the same decomposition that makes the task-level CV caveat above a re-projection onto a shifted location rather than a contradiction (the effective median dilates under coupling, Section 5).

Limitations. All results are numerical (Digital-Model level; no fabricated device); the task-level CV sweep reveals an operating-regime-dependent optimum (Supplementary Note 1); the homeostat's gains are substrate-level (task-level NMSE is readout-compensated); FTLE estimates are finite-time and resolution-limited near zero.

8 Conclusion

A physics-constrained relaxation substrate with per-pulse contrast coupling possesses a well-defined memory–chaos phase diagram: negative-feedback coupling buys 24–53% more held-out memory up to a sharp transition, positive-feedback coupling self-limits at criticality with maximal separation, chaos destroys linear memory, and the log-normal trap spectrum fixes a forgetting kernel that predicts the measured memory curve ($r = 0.97$) with a fabrication-tunable tail. A λ-homeostat restores 7.9–18% of memory after disturbances. These results turn a device-physics spread parameter into a computational design knob and provide the substrate-level theory for the REDEM online-learning architecture (companion paper [29]).

A Derivation details

A.1 The trap spectrum is log-normal (Section 2)

A shallow trap escapes by thermally activated emission with rate $\nu e^{-E/(kT)}$, so a single trap's time constant is $\tau(E) = \nu^{-1}e^{E/(kT)}$. With median activation energy E_a, $\tau_0 = \nu^{-1}e^{E_a/(kT)}$; for $E_a = 0.55$ eV, $\nu = 10^{13}$ s^{-1}, $T = 300$ K ($kT \approx 25.85$ meV): $\tau_0 = 10^{-13}e^{0.55/0.02585}\,\text{s} \approx 174\,\mu\text{s}$. If the activation energy varies across devices as $E_i = E_a + \Delta E_i$ with $\Delta E_i \sim \mathcal{N}(0, \sigma_E^2)$, then

$$\ln \tau_i = \ln \nu^{-1} + (E_a + \Delta E_i)/(kT) \sim \mathcal{N}\big(\ln \tau_0,\ \sigma_E^2/(kT)^2\big), \tag{A.1}$$

i.e., τ_i is log-normal with $\sigma^2 = \sigma_E^2/(kT)^2$ and $\mathrm{CV}^2 = e^{\sigma^2} - 1$. The CV = 0.20 used throughout corresponds to a spread $\sigma_E = kT\sqrt{\ln(1 + 0.20^2)} \approx 5.1$ meV — a few-meV fabrication spread — which motivates treating CV as a physically plausible design knob rather than a free hyperparameter.

A.2 Per-pulse update and the order independence of coupling (Section 2)

Over a write pulse of width p_w, the injection term drives occupancy toward 1 at rate $\alpha_{\text{eff},i}$; over the inter-pulse interval Δt_t both the injected and the pre-existing occupancy relax with the trap time constant. The discrete per-pulse map (Eq. (1)) is the composition of one injection step and one relaxation step:

$$x_i(t+1) = \Big[x_i(t) + \alpha_{\text{eff},i}(t)\big(1 - x_i(t)\big)\Big]\, e^{-(p_w + \Delta t_t)/\tau_i}. \tag{A.2}$$

Two-pass order independence: the contrast $g_i(t)$ is evaluated from the *pre-update* state $x(t)$ alone (neighbor current ratios at time t), and all N units are then updated simultaneously from the same pre-update state. The map is therefore a well-defined function of the state at t — a parallel map — and the order in which individual unit updates are computed is irrelevant. At $\kappa = 0$, $\alpha_{\text{eff},i} = \alpha_0$ for all i and the coupling term vanishes; the map reduces exactly to the parallel array of [4] (verified to $\leq 10^{-10}$ in the implementation's self-test).

A.3 The forgetting kernel and its quadrature (Section 5)

For a single trap the linear memory of an input at lag t decays as $e^{-t/\tau}$; averaging over the log-normal spectrum gives Eq. (3). Substituting $z = (\ln\tau - \mu)/(\sqrt{2}\sigma)$, with $d\tau = \sqrt{2}\,\sigma e^{\mu+\sqrt{2}\sigma z}dz$ and $p(\tau)d\tau = \pi^{-1/2}e^{-z^2}dz$:

$$M(t) = \pi^{-1/2}\int_{-\infty}^{\infty} e^{-z^2}\exp\big(-t\,e^{-(\mu+\sqrt{2}\sigma z)}\big)\,dz \;\approx\; \pi^{-1/2}\sum_{i=1}^{n} w_i\,\exp\big(-t\,e^{-(\mu+\sqrt{2}\sigma z_i)}\big), \tag{A.3}$$

where (z_i, w_i) are the n-point Gauss–Hermite nodes and weights ($n = 160$ used; the quadrature is exact to machine precision for t up to 10^4 pulses).

Median pinning of the $1/e$ horizon. The integrand peaks at $\tau^*(t)$ where the saddle-point condition $t/\tau^* = 1 + (\ln\tau^* - \mu)/\sigma^2$ holds (from $d/d\tau\,[\ln p(\tau) - t/\tau] = 0$). At $t = \tau_0$ the solution is $\tau^* = \tau_0$ up to a weak CV-dependent correction, so $M(\tau_0) \approx e^{-1}$: the $1/e$ horizon is pinned at $\approx \tau_0/\bar{\Delta}t \approx 16$ pulses regardless of CV (numerically 12–16 pulses for CV $\in [0.02, 1.0]$, Fig. 3).

Tail asymptotics. For $t \gg \tau_0$ the saddle point moves to $\tau^* \approx t\,\sigma^2/\ln(t/\tau_0)$, and $\ln M(t) \approx -(\ln\tau^* - \mu)^2/(2\sigma^2) - t/\tau^*$; dropping the t/τ^* term gives the leading-order log–log slope

$$\frac{d\ln M}{d\ln t} \approx -\frac{\ln t - \mu}{\sigma^2}. \tag{A.4}$$

The t/τ^* correction is not sub-leading in the sampled window: over $t \in [5{,}000,\ 20{,}000]$ pulses the exact quadrature slope is -60.6 at CV $= 0.2$ ($\sigma^2 = 0.0392$; $\mu = \ln(\tau_0/\bar{\Delta}t) \approx 2.76$ in pulse units), while Eq. (A.4) gives -147 to -182 — the asymptotic regime is only partially reached (at $t = 200$ the exact slope is already -9.3, against the formula's -64.8). The qualitative prediction is confirmed: the slope steepens as the spectrum narrows (quadrature -60.6 at CV $= 0.2$ vs. -7.1 at CV $= 1.0$; Eq. (A.4): -147 to -182 vs. -8.3 to -10.3), and CV$\to 0$ makes $\sigma^2 \to 0$ and the slope $\to -\infty$: the single-exponential limit.

A.4 Benettin algorithm (Section 3)

1. Evolve the main trajectory $x(t)$ under the fixed drive stream for the full collection window.
2. At each renormalization point $t_m = m\,T_{\text{ren}}$ ($T_{\text{ren}} = 10$ pulses), create the twin $\tilde{x}(t_m) = x(t_m) + \delta$ with $\delta_i \sim U(-\epsilon, \epsilon)$ per unit, $\epsilon = 10^{-8}$.
3. Evolve $\tilde{x}$ in parallel with x under the *identical* drive sequence for T_{ren} pulses.
4. Measure the separation $d_m = \|\tilde{x}(t_{m+1}) - x(t_{m+1})\|_2$, accumulate $\ln(d_m/(\epsilon\sqrt{N}))$, and renormalize the twin onto the reference direction: $\tilde{x} \leftarrow x + (\epsilon\sqrt{N})(\tilde{x} - x)/d_m$.
5. After M renormalization blocks, $\lambda \approx (MT_{\text{ren}})^{-1}\sum_{m=1}^{M}\ln(d_m/(\epsilon\sqrt{N}))$ per pulse.

The estimate is finite-time and resolution-limited near zero ($|\lambda| \lesssim 10^{-3}$ is indistinguishable from 0); the perturbation ϵ is small enough that the linearized dynamics dominates within one block.

A.5 Memory-capacity estimator and the linear-reservoir closed form (Section 3)

For lag k the target is $y_k(t) = \Delta t_{t-k}$ and the features are the standardized current ratios plus bias, ϕ_t. Ridge regression on the first 70% of the stream solves $W_k = (\Phi^\top\Phi + \lambda_r I)^{-1}\Phi^\top Y_k$ (targets mean-centered); evaluation is on the last 30% after a $k_{\max} = 50$-step leakage buffer, so no evaluation target references training-segment inputs. The per-lag capacity is the squared multiple correlation $\mathrm{MC}(k) = r_k^2 = \mathrm{corr}(y_k, \hat{y}_k)^2$, $\mathrm{MC}_{\mathrm{total}} = \sum_{k=1}^{50} r_k^2$.

Linear closed form. Linearize the substrate (clip inactive, κ small): $x_{t+1} = Ax_t + bu_{t+1}$ with $u_t = \Delta t_t$ and A contractive. Then $x_t = \sum_{j\geq 0} A^j b\, u_{t-j}$; the input at lag k contributes the component $A^k b\, u_{t-k}$, whose covariance with x_t is $c_k = A^k b\, \mathrm{Var}(u)$. The best linear predictor achieves the squared multiple correlation

$$\mathrm{MC}(k) = \frac{c_k^\top \Sigma_x^{-1} c_k}{\mathrm{Var}(u)}, \qquad \Sigma_x = \mathrm{Var}(x_t) = \sum_{j\geq 0} A^j bb^\top (A^\top)^j\, \mathrm{Var}(u), \tag{A.5}$$

where the division by the target variance $\mathrm{Var}(u) = \mathrm{Var}(y_k)$ is required for dimensional consistency: c_k carries the units of $x\, u$, so $c_k^\top \Sigma_x^{-1} c_k$ alone is the *explained variance* in units of u^2, and only after normalization by $\mathrm{Var}(u)$ does $\mathrm{MC}(k)$ become the dimensionless squared correlation used throughout. Dambre et al. [20] show the total linear capacity of an N-dimensional linear system is bounded by the state dimension: $\sum_k \mathrm{MC}(k) \leq N$, with equality attained by orthogonally normalized dynamics. The measured nonlinear-substrate curve is estimated with the held-out protocol of Section 3; the closed form anchors the theory at the linear limit.

Supplementary Material

Supplementary Note 1. Task-level CV sweep (Fig. S1). At the task level (held-out MC at the memory-optimal operating points, $\mathrm{CV} \in \{0.1, 0.2, 0.4\}$, 10 seeds), the uncoupled substrate follows the single-unit kernel theory: MC rises $2.54 \pm 0.26 \to 3.37 \pm 0.33$ (+33%) as CV widens from 0.1 to 0.4 — the heavier tail retains more old lags. The coupled substrate at the near-critical optimum behaves *oppositely*: random_graph $\kappa = 25$ MC falls $12.93 \pm 0.62 \to 7.33 \pm 1.33$ as CV widens (narrow spectrum best), and ring_bidir $\kappa = 20$ falls $9.55 \pm 0.60 \to 7.86 \pm 1.85$. The collective near-critical state prefers homogeneous timescales — heterogeneous relaxation rates disrupt the synchronization of the contrast-feedback loop. The CV knob's direction therefore depends on the operating regime: wide spectra for parallel (kernel-dominated) operation, narrow spectra for coupled near-critical operation.

Declaration of competing interest

The author declares no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Author contributions (CRediT)

Yaming Hu: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Data curation, Writing – original draft, Writing – review & editing, Visualization.

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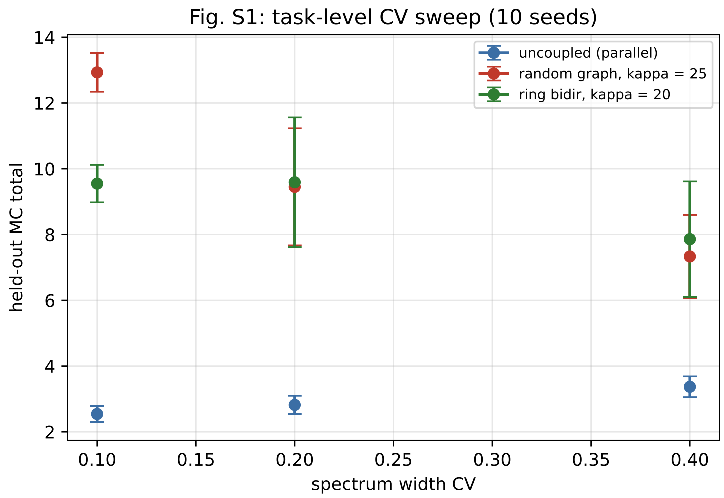

Figure S1: Task-level CV sweep (held-out MC vs. CV, 10 seeds; data/s10_cv_sweep_v1.csv).

Code and data availability

All simulation scripts and generated data are available at `https://github.com/huyamingc/REDEM` (private during review; released on acceptance).

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